

# Requirements Engineering

## Overview

**Business Requirements** represent a **high-level objective** or **goal** of the organization or customer, it describes

**why the organization is implementing** the system.

## Analyst Questions

## Executive Responses

What motivates your interest in a chemical tracking system?

Managing chemical inventories manually costs too much and is inefficient.

How much would you like to reduce your chemical expenses?

By 25% within one year.

What is keeping you from cutting by 25% today?  
What is causing the high cost and inefficiency?

We buy unnecessary chemicals because we don't know what we have in inventory. We discard too much unused material that has expired.

Anything else I should know?

Placing orders is complicated; it takes users a long time. The government reports we create are manually generated, which takes far too much time.

**FIGURE 5-4** Example of a conversation between a business analyst and an executive sponsor.

# Vision Statement

- **For** [target customer]
- **Who** [statement of the need or opportunity]
- **The** [product name]
- **Is** [product category]
- **That** [major capabilities, key benefit, compelling reason to buy or use]
- **Unlike** [primary competitive alternative, current system, current business process]
- **Our product** [statement of primary differentiation and advantages of new product]

# Vision Statement

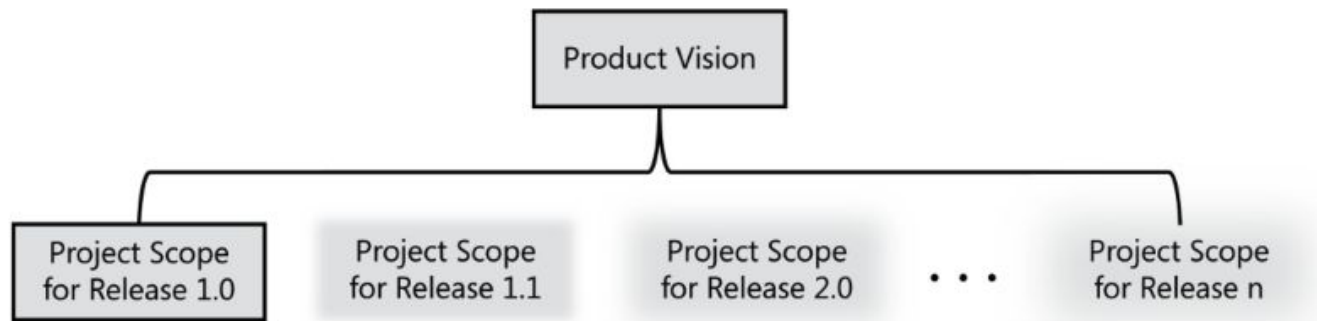
*For scientists **who** need to request containers of chemicals, **the** Chemical Tracking System **is** an information system **that** will provide a single point of access to the chemical stockroom and to vendors. The system will store the location of every chemical container within the company, the quantity of material remaining in it, and the complete history of each container's locations and usage. This system will save the company 25 percent on chemical costs in the first year of use by allowing the company to fully exploit chemicals that are already available within the company, dispose of fewer partially used or expired containers, and use a standard chemical purchasing process. **Unlike** the current manual ordering processes, **our product** will generate all reports required to comply with federal and state government regulations that require the reporting of chemical usage, storage, and disposal.*

The scope description establishes the **boundary and connections** between the system you're developing and everything else in the universe.

The **context diagram** visually illustrates this boundary.

# Vision and Scope

- The vision applies to the product **as a whole**.
- The scope pertains to a **specific project or iteration** that will implement the next increment of the product's functionality.



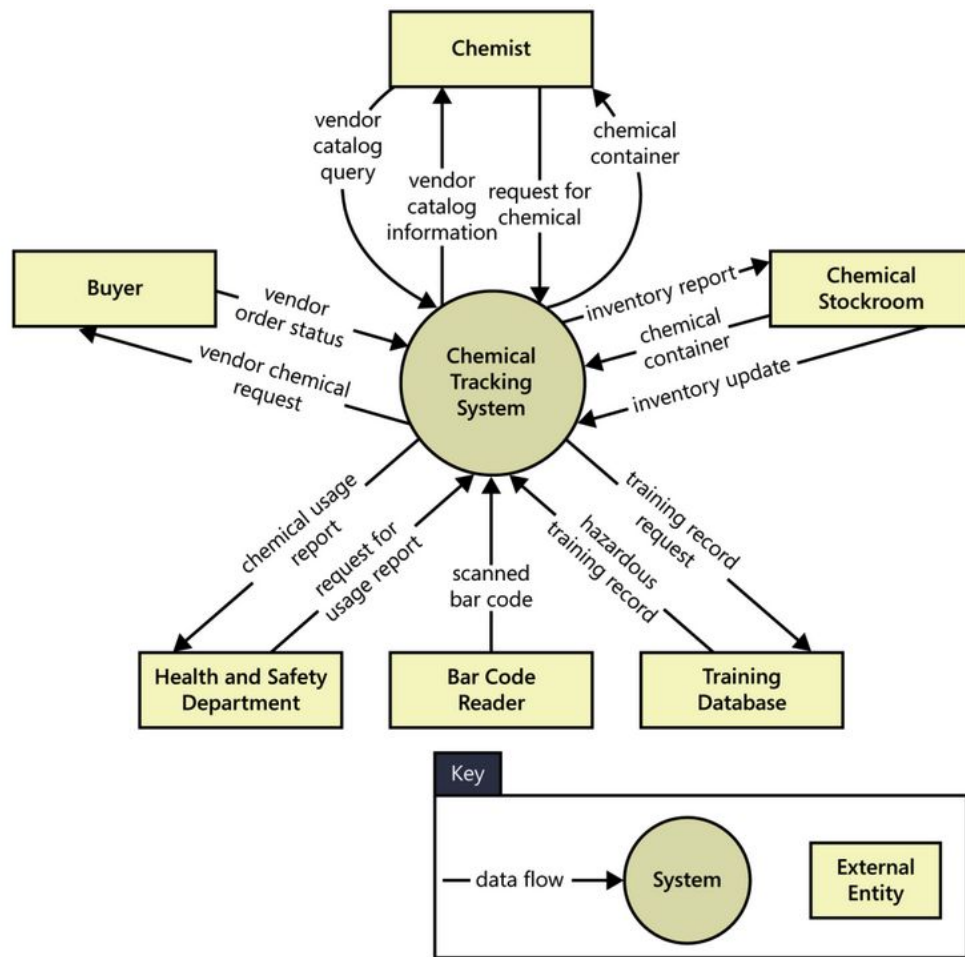
**FIGURE 5-1** The product vision encompasses the scope for each planned release, which is less well defined the farther out you look.

# Scope Representation Techniques

The models described in this section can be used to represent project scope in various ways:

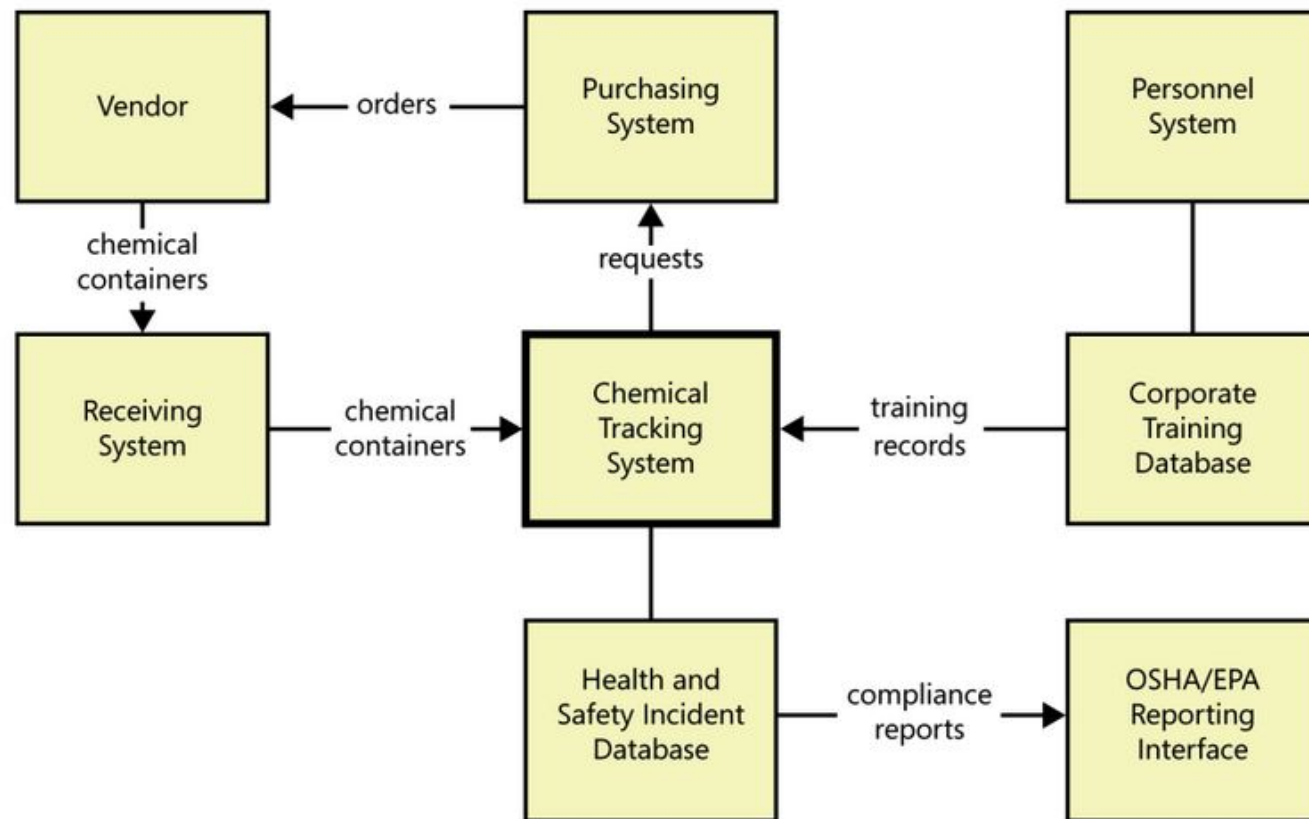
- 1) **Context diagram**
- 2) Ecosystem Map
- 3) Feature Tree
- 4) Even List





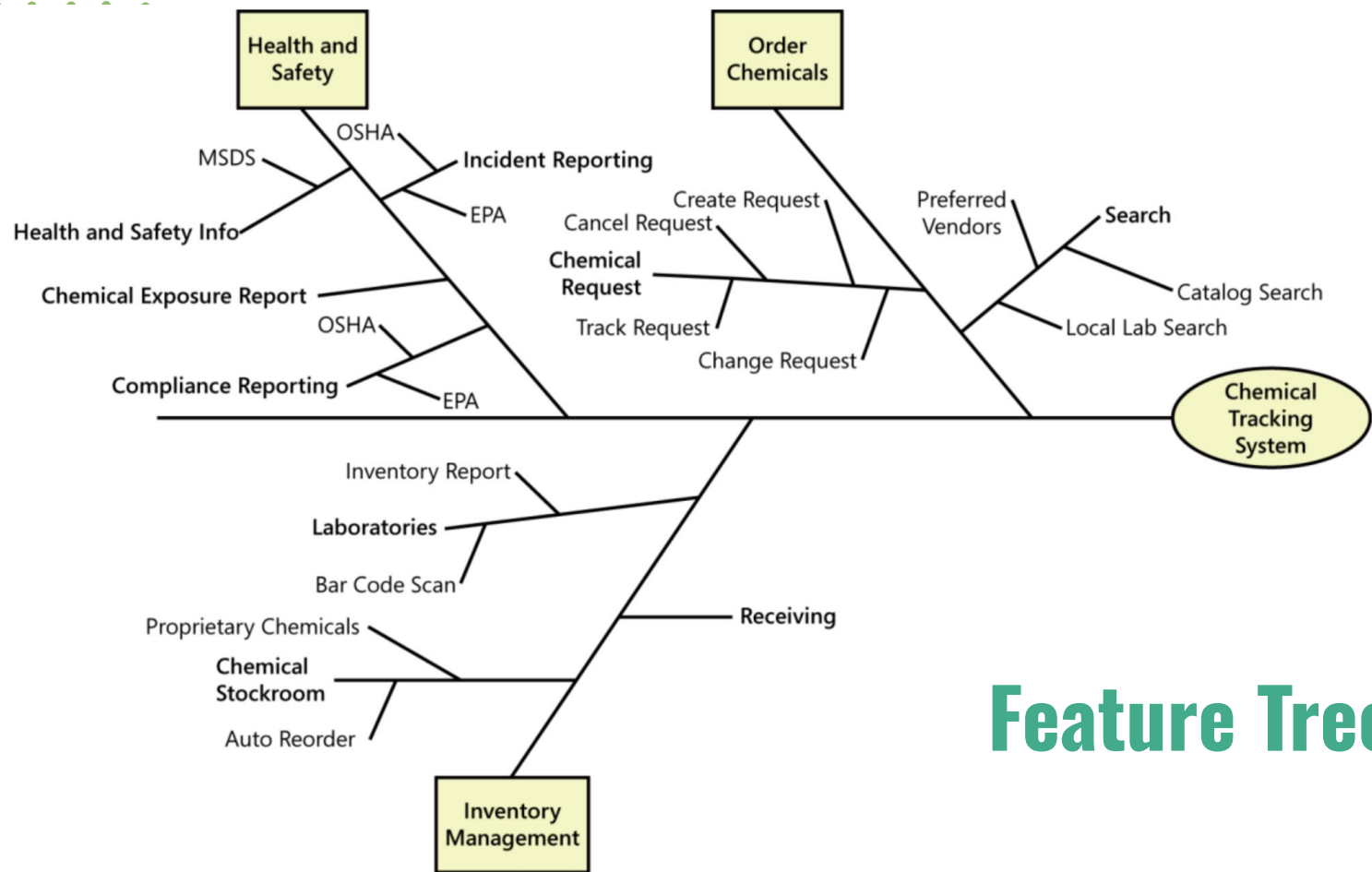
# Context Diagram

**FIGURE 5-6** Partial context diagram for the Chemical Tracking System.



# Ecosystem Map

**FIGURE 5-7** Partial ecosystem map for the Chemical Tracking System.



## Feature Tree

**FIGURE 5-8** Partial feature tree for the Chemical Tracking System.

## External Events for Chemical Tracking System

- Chemist places a chemical request.
- Chemical container bar code is scanned.
- Time to generate OSHA compliance report arrives.
- Vendor issues new chemical catalog.
- New proprietary chemical is accessioned into system.
- Vendor indicates chemical is backordered.
- Chemist asks to generate his chemical exposure report.
- Updated material safety datasheet is received from EPA.
- New vendor is added to preferred vendor list.
- Chemical container is received from vendor.

**FIGURE 5-9** Partial event list for the Chemical Tracking System.

# Event List

# User Classes

**TABLE 6-1** User classes for the Chemical Tracking System

| Name                                         | Number                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemists (favored)                           | Approximately 1,000 located in 6 buildings | Chemists will request chemicals from vendors and from the chemical stockroom. Each chemist will use the system several times per day, mainly for requesting chemicals and tracking chemical containers into and out of the laboratory. The chemists need to search vendor catalogs for specific chemical structures imported from the tools they use for drawing structures.                                                                                   |
| Buyers                                       | 5                                          | Buyers in the purchasing department process chemical requests. They place and track orders with external vendors. They know little about chemistry and need simple query facilities to search vendor catalogs. Buyers will not use the system's container-tracking features. Each buyer will use the system an average of 25 times per day.                                                                                                                    |
| Chemical stockroom staff                     | 6 technicians, 1 supervisor                | The chemical stockroom staff manages an inventory of more than 500,000 chemical containers. They will supply containers from three stockrooms, request new chemicals from vendors, and track the movement of all containers into and out of the stockrooms. They are the only users of the inventory-reporting feature. Because of their high transaction volume, features that are used only by the chemical stockroom staff must be automated and efficient. |
| Health and Safety Department staff (favored) | 1 manager                                  | The Health and Safety Department staff will use the system only to generate predefined quarterly reports that comply with federal and state chemical usage and disposal reporting regulations. The Health and Safety Department manager will request changes in the reports periodically as government regulations change. These report changes are of the highest priority, and implementation will be time critical.                                         |

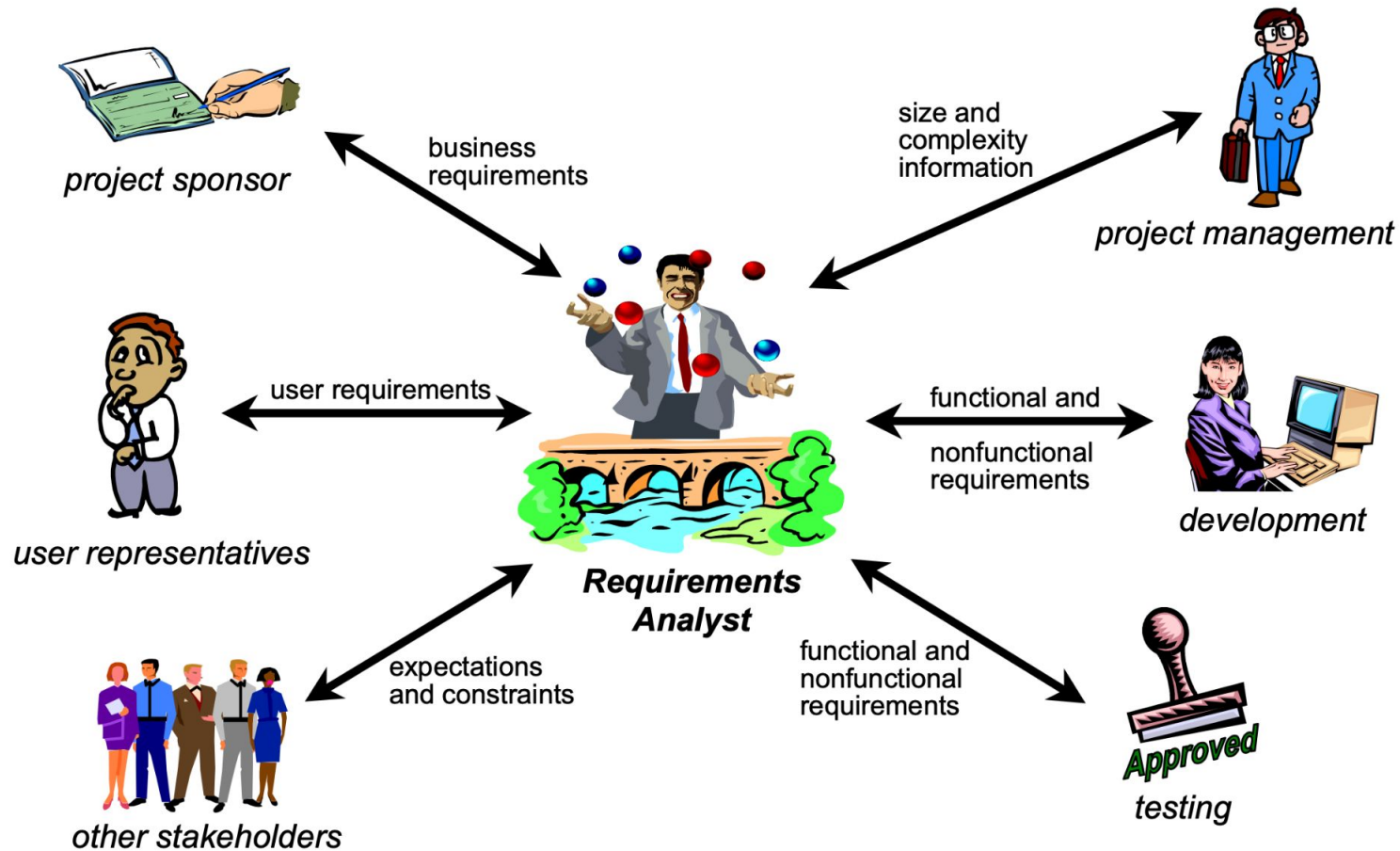
# User Personas

A **persona is a description**  
of a hypothetical,  
generic person who serves as a  
**stand-in for a group of users**  
having similar characteristics  
and needs.

Here's an example of a persona for one user class on the Chemical Tracking System:

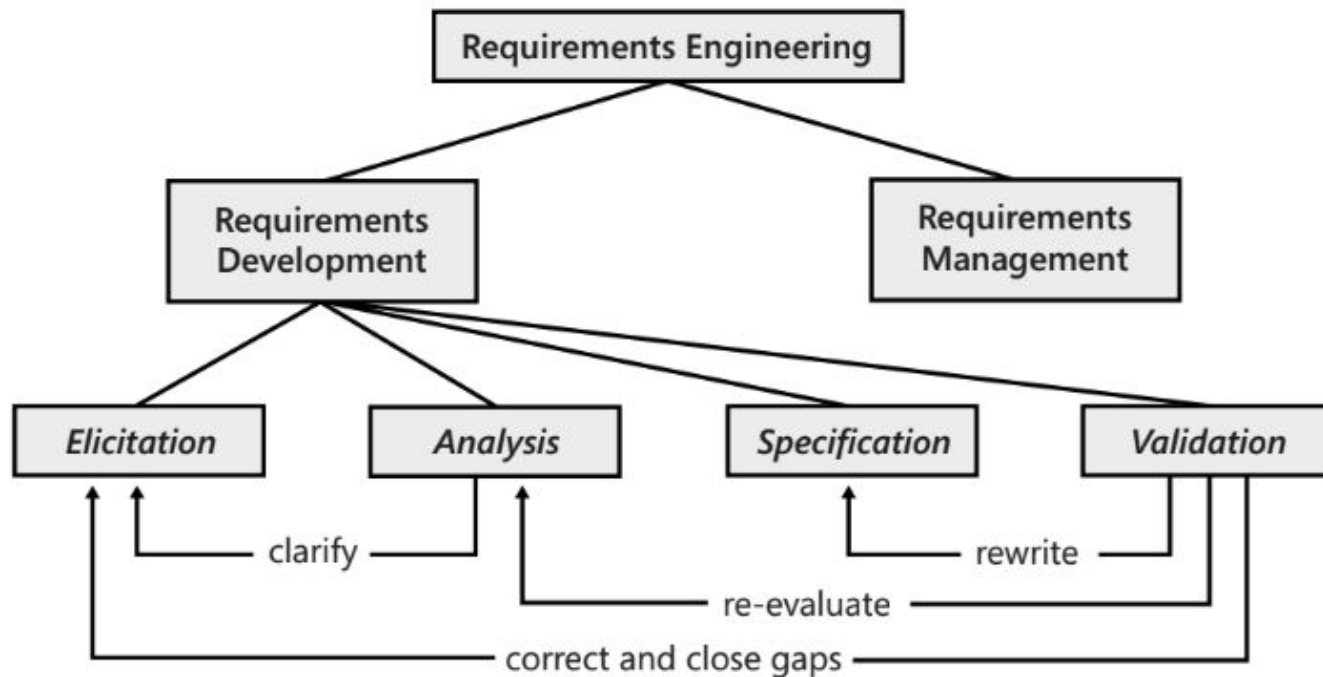
*Fred, 41, has been a chemist at Contoso Pharmaceuticals since he received his Ph.D. 14 years ago. He doesn't have much patience with computers. Fred usually works on two projects at a time in related chemical areas. His lab contains approximately 300 bottles of chemicals and gas cylinders. On an average day, he'll need four new chemicals from the stockroom. Two of these will be commercial chemicals in stock, one will need to be ordered, and one will come from the supply of proprietary Contoso chemical samples. On occasion, Fred will need a hazardous chemical that requires special training for safe handling. When he buys a chemical for the first time, Fred wants the material safety data sheet emailed to him automatically. Each year, Fred will synthesize about 20 new proprietary chemicals to go into the stockroom. Fred wants a report of his chemical usage for the previous month to be generated automatically and sent to him by email so that he can monitor his chemical exposure.*







# Requirements Engineering Activities



# Cosmic Truths About Software Requirements

Requirements **Realities**

## Cosmic Truth #1:

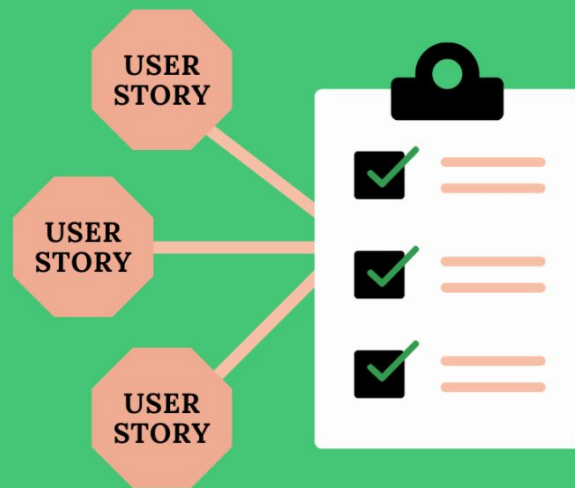
If you don't get the requirements right, it doesn't matter how well you execute the rest of the project.



## ACCEPTANCE CRITERIA

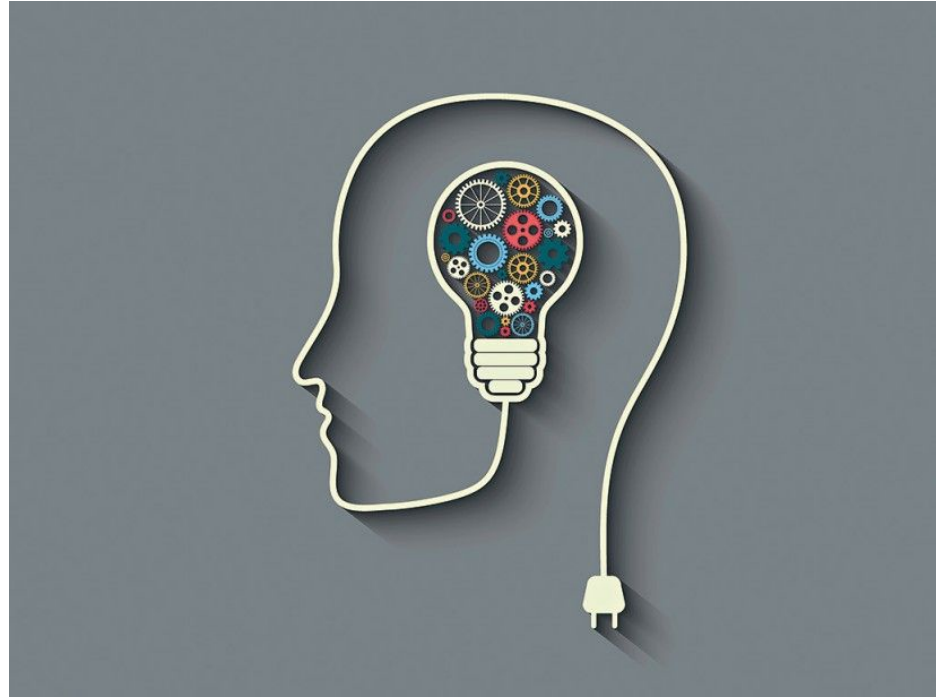


## DEFINITION OF DONE



## Cosmic Truth #2:

Requirements development is a discovery and invention process, not just a collection process.



## Cosmic Truth #3:

Change happens.

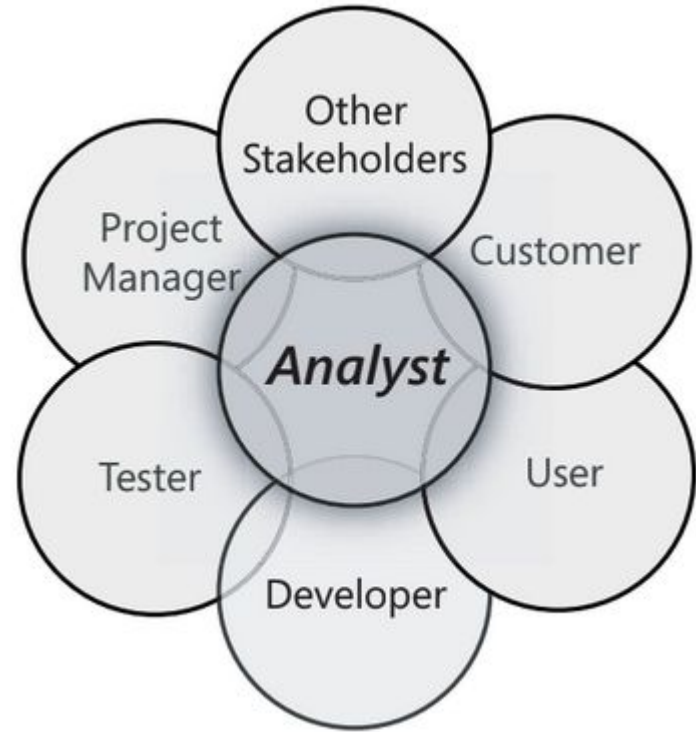


# Cosmic Truths About Software Requirements

Requirements **Stakeholders**

## Cosmic Truth #4:

The interests of all the project stakeholders intersect in the requirements process.





## Cosmic Truth #5:

Customer involvement is the most critical contributor to software quality.



## Cosmic Truth #5:

**Identify user classes.** Unrepresented user classes are likely to be disappointed with the project outcome.

**Select product champions.** Product champions are actual users who represent their user-class peers.

**Build prototypes.** Prototypes are far more tangible than written requirements specifications. However, prototypes aren't a substitute for documenting the detailed requirements.

**Agree on customer rights and responsibilities.**

## Cosmic Truth #6:

The customer is not always right, but the customer always has a point.

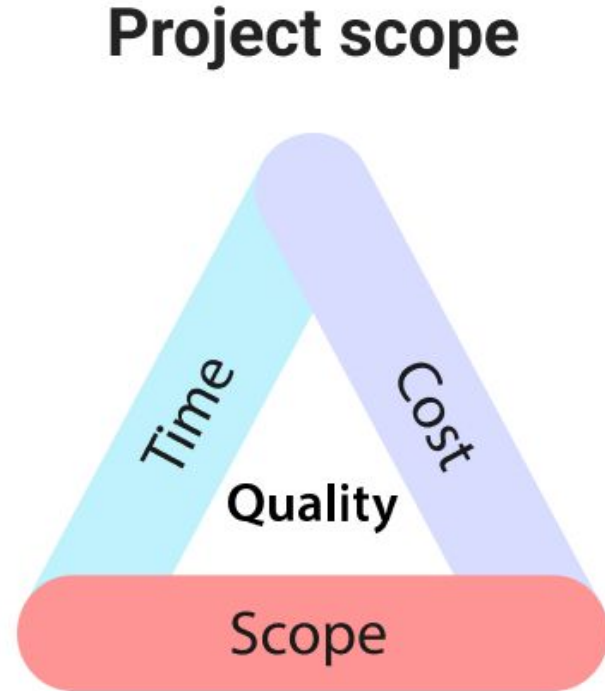


# Cosmic Truths About Software Requirements

Requirements **Specifications**

## Cosmic Truth #7:

The first question an analysis should ask about a proposed new requirement is, “Is this requirement in scope?”



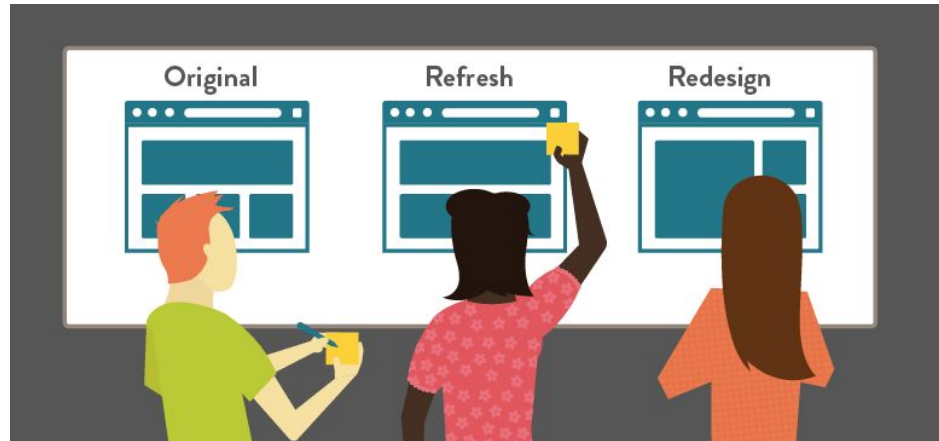
## Cosmic Truth #8:

Even the best requirements document cannot—and should not—replace human dialogue.



## Cosmic Truth #9:

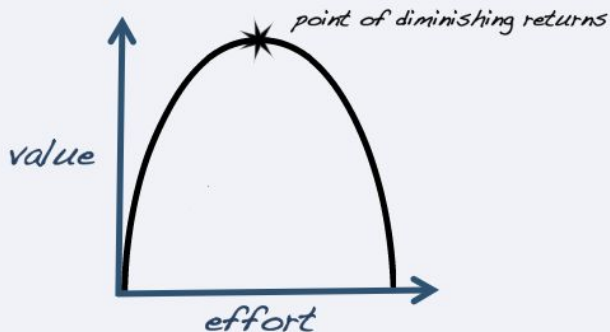
The requirements might be vague, but the product will be specific.



## Cosmic Truth #10:

You're never going to have perfect requirements.

### “GOOD ENOUGH”

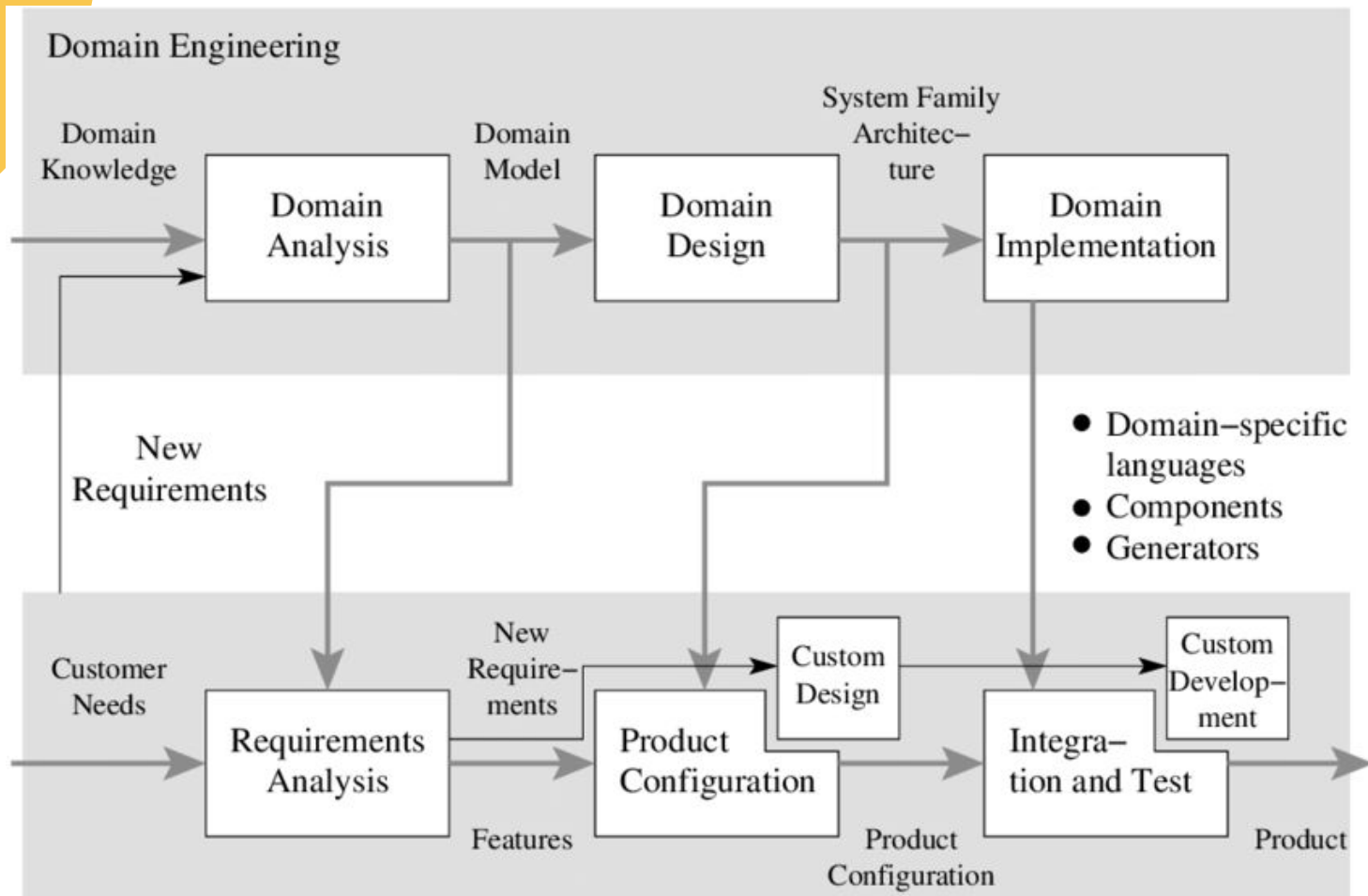




# Application domain

An application domain is **the segment of reality for which a software system is developed**. It is the background or starting point for the actual-state analysis and the creation of a domain model.

An application domain can be an organization, a department within an organization, or a single workplace.

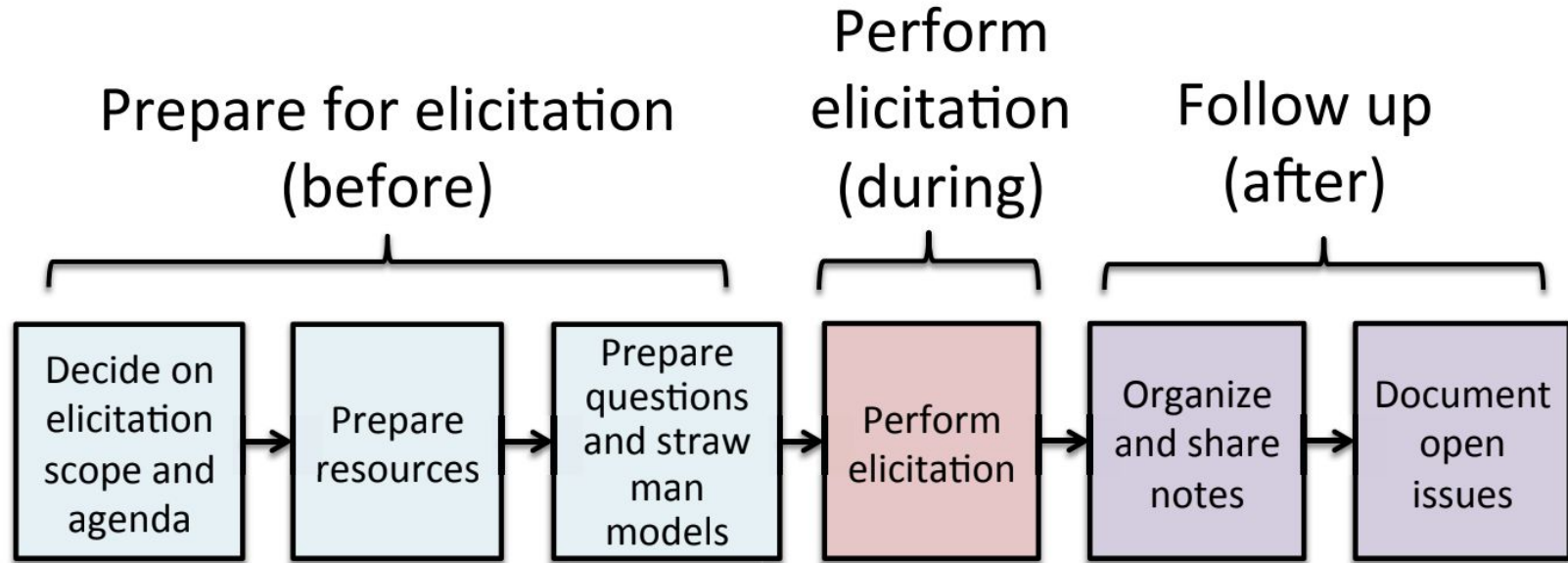


# Elicitation Techniques

Some popular techniques are:

- a) Interviews
- b) Questionnaires
- c) Workshops
- d) Brainstorming
- e) Focus Group
- f) Introspection
- g) Task Analysis
- h) Card Sorting
- i) Scenarios
- j) Viewpoints
- k) Prototyping
- l) System interface analysis
- m) User interface analysis
- n) Ethnography
- o) Apprenticing
- p) Observations
- q) Document analysis
- r) Domain Analysis

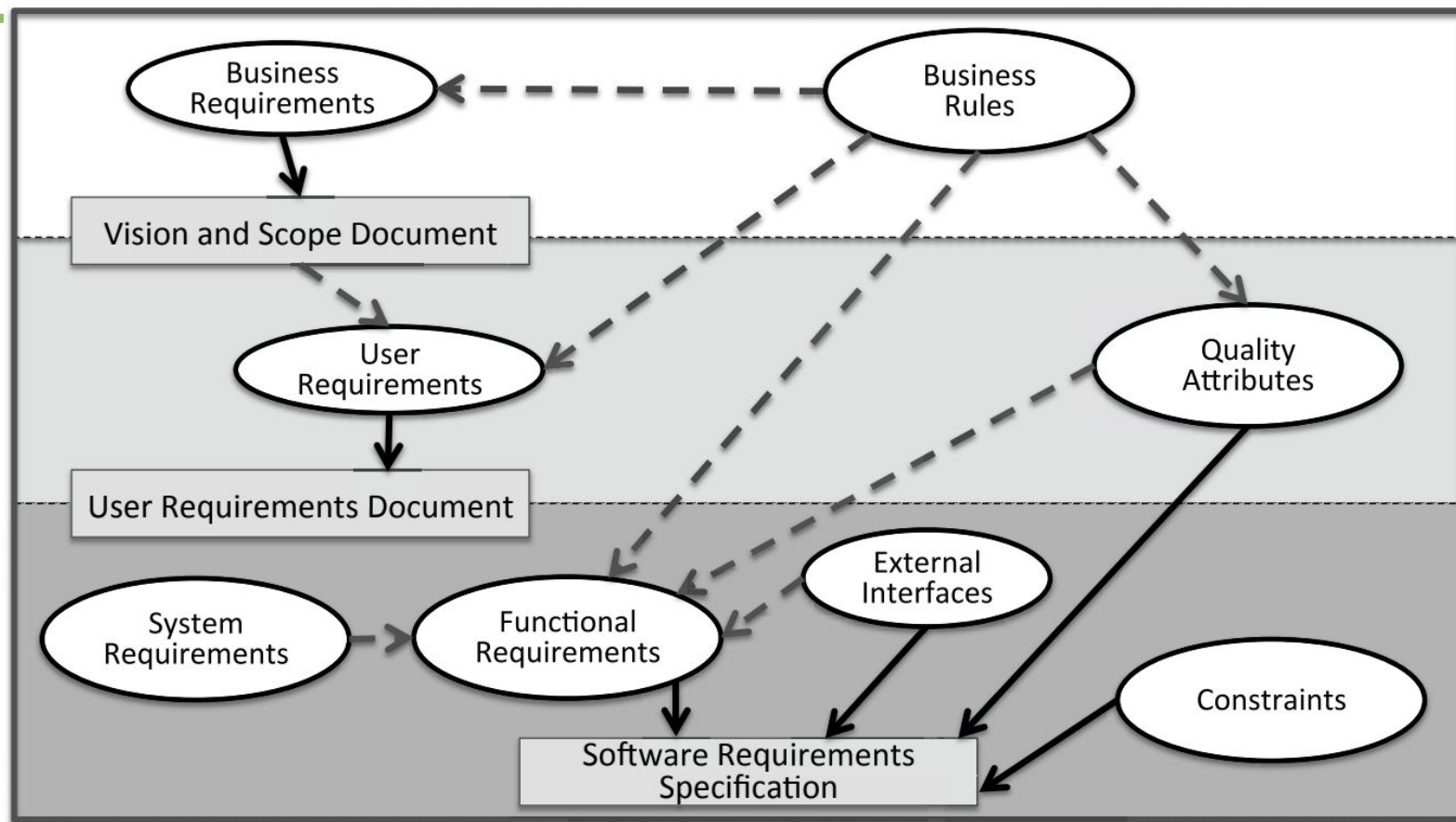
# Single Elicitation Session

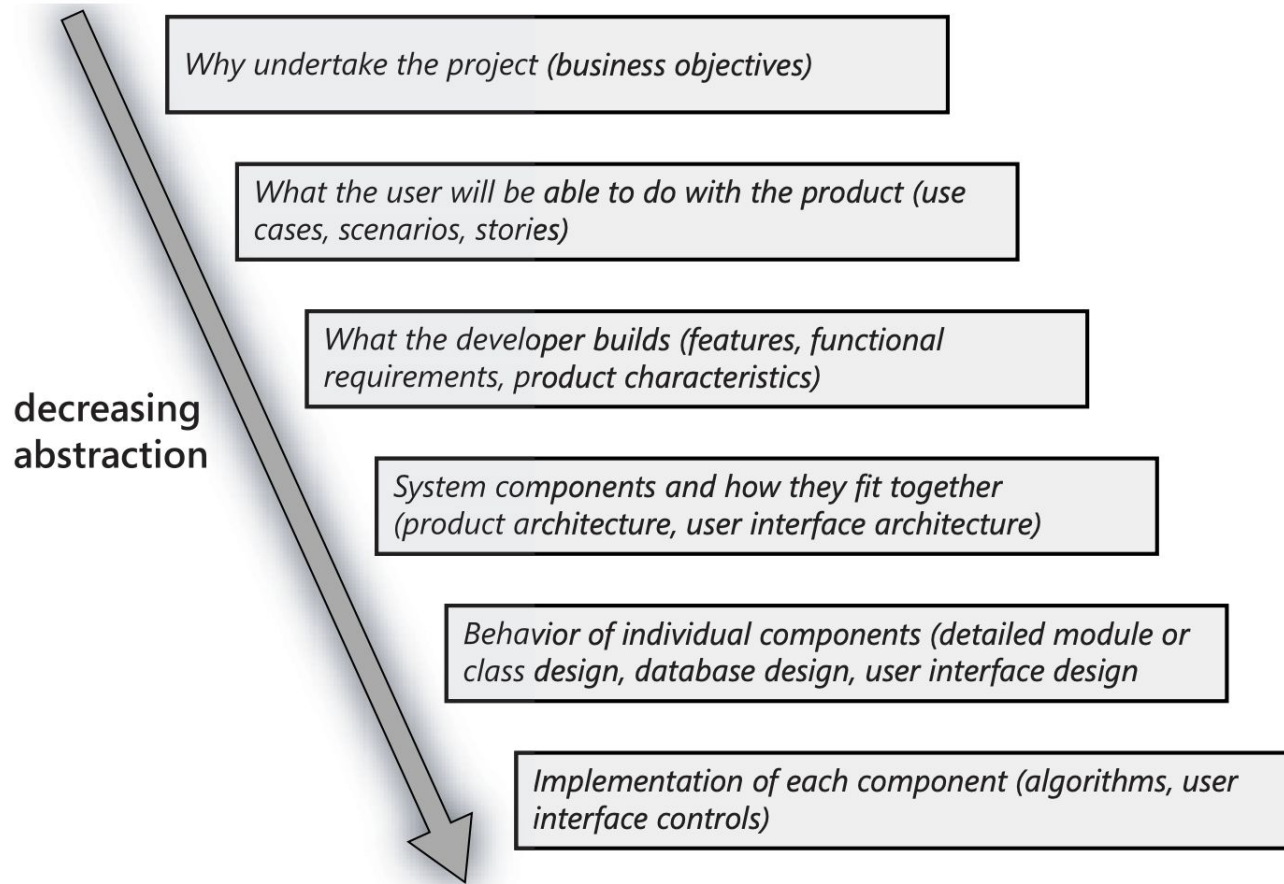


Elicitation Activities

# Priority of Requirements







**Figure 16-1** Levels of abstraction in software specification and development.

# User Stories

User stories are composed of three aspects:

1. A **written description** of the story used for planning and as a reminder.
2. **Conversations** about the story that serve to flesh out the details of the story.
3. **Tests** that convey and document details and that can be used to determine when a story is complete.



# User Story Description

Good:

- As a business traveler, I want to cancel a reservation ...
- As a vacation traveler, I want to see photos of the hotels
- ...

Bad:

- As a manager, I want the system to be stable so that I can please my customers ...
- As a manager, I want the system to be written in Ruby ...

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# User Story Conversations

When a story is too large, it is sometimes referred to as an **epic**.

Epic: A user can search for a job.

- A user can search for jobs by attributes like location, salary range, job title, company name, and the date the job was posted.
- A user can view detailed information about a company that has posted a job.
- ...



Conversations

# User Stories

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# User Story Test

A user can search for jobs by attributes like location, salary range, job title, company name, and the date the job was posted.

- Try it with an empty job description.
- Try it with a really long job description.
- Try it with a missing salary.
- Try it with a six-digit salary.

} Tests

## One alternative:

“As a business traveller I want to cancel a reservation”

**Front**

- ☐ A premium member can cancel the same day without fee.
- ☐ A premium member is charged 10% for same day cancellations
- ☐ An email confirmation is sent
- ☐ Notify the hotel about the cancellation

**Back**

Conditions of satisfaction  
Acceptance criteria



## Sistema de Gestión de Emergencias Médicas Inteligente (SGEMI)

La atención de **emergencias médicas** es un proceso crítico que involucra múltiples actores, decisiones urgentes y recursos limitados, donde una mala coordinación puede generar retrasos, saturación de servicios y riesgos graves para los pacientes. El Sistema de Gestión de Emergencias Médicas Inteligente (SGEMI) es una plataforma digital diseñada para apoyar la **gestión, coordinación y supervisión del proceso** completo de atención de emergencias médicas, desde la recepción del reporte inicial hasta la conclusión del servicio médico. El proceso de atención inicia cuando una emergencia médica es reportada al centro de atención de emergencias. Dicho reporte puede provenir de distintas fuentes, como ciudadanos, personal médico, instituciones públicas o sistemas externos.

Una vez recibido el reporte, este es atendido por un operador del centro de emergencias, quien es responsable de registrar la información inicial del incidente y clasificar preliminarmente la emergencia con base en protocolos definidos. A partir de esta información, el sistema deberá permitir que personal autorizado del **centro de emergencias evalúe la situación y asigne una prioridad**, considerando factores como la gravedad del incidente, la ubicación, la disponibilidad de recursos y las condiciones operativas actuales. Esta priorización será utilizada para decidir la asignación de recursos, como ambulancias y personal médico, de manera eficiente y oportuna. Una vez asignados los recursos, el sistema dará soporte al seguimiento del servicio, permitiendo monitorear el estado de la ambulancia, el avance del traslado y cualquier cambio relevante en la condición del paciente. Durante este proceso, el sistema deberá contemplar que las condiciones pueden cambiar, como la indisponibilidad de hospitales, el deterioro del paciente o la necesidad de reasignar recursos.



El sistema también permitirá la coordinación con hospitales y centros de atención médica, proporcionando visibilidad sobre su capacidad de recepción, disponibilidad de personal y estado operativo, con el fin de apoyar decisiones informadas durante el traslado y la atención del paciente.

Una vez concluida la atención de la emergencia, el sistema registrará la información final del servicio, permitiendo su consulta posterior para fines de análisis, supervisión y mejora continua. Esta información histórica será utilizada por personal administrativo y de gestión para evaluar tiempos de respuesta, uso de recursos, tipos de incidentes y desempeño general del sistema.

SGEMI está diseñado como un sistema evolutivo, capaz de incorporar nuevas funcionalidades, reglas de negocio o integraciones conforme cambien las necesidades operativas, tecnológicas o administrativas del servicio de emergencias médicas.

El sistema no busca reemplazar el criterio humano, sino proveer información estructurada, oportuna y confiable que facilite la toma de decisiones en un entorno de alta presión, donde la coordinación efectiva es fundamental.